L.T. of derivative functions.

If $\mathcal{L}[f(t)] = F(s)$

and $f'(t)$ is a derivative, then,

$$\mathcal{L}[f'(t)] = s\mathcal{L}[f(t)] = f(0)$$

Proof,

$$\mathcal{L}[f'(t)] = \int_0^\infty e^{-st} f'(t)\, dt$$

$$= \left[e^{-st} f(t)\right]_0^\infty - \int_0^\infty f(t) \cdot -se^{-st}\, dt$$

$$= [0 - f(0)] + s\int_0^\infty e^{-st} f(t)\, dt$$

$$\therefore \mathcal{L}[f'(t)] = s \cdot \mathcal{L}[f(t)] - f(0)$$

Similarly,

$$\mathcal{L}[f''(t)] = s^2 \cdot \mathcal{L}[f(t)] - s \cdot f(0) - f'(0)$$

$$\mathcal{L}[f'''(t)] = s^3 \cdot \mathcal{L}[f(t)] - s^2 f(0) - sf'(0) - f''(0)$$

L.T. of integral functions

If $\mathcal{L}[f(t)] = F(s)$ then,

$$\mathcal{L}\left[\int_0^t f(t)\, dt\right] = \frac{1}{s} F(s)$$

Prove the following,

Q. $\mathcal{L}(1) = \frac{1}{s}$

A. $\mathcal{L}(1) = \int_0^\infty e^{-st} \cdot 1 \, dt$

$= \left[\frac{e^{-st}}{-s}\right]_0^\infty = \frac{-1}{s}[0-1] = \frac{1}{s}$

$\mathcal{L}(af(t)) = a\mathcal{L}(f(t))$

Q. $\mathcal{L}(e^{at}) = \frac{1}{s-a}$, $s > a$

A.

$= \int_0^\infty e^{-st} \cdot e^{at} \, dt = \int_0^\infty e^{-(s-a)t} \, dt$

$= \left[\frac{e^{-(s-a)t}}{-(s-a)}\right]_0^\infty = \frac{-1}{s-a}[0-1]$

$= \frac{1}{s-a}$

Q. $\mathcal{L}[\sin at] = \frac{a}{s^2+a^2}$

A. $= \int_0^\infty e^{-st} \sin at \, dt$, we know $\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$

$\therefore \mathcal{L}(\sin at) = \mathcal{L}\left(\frac{e^{iat} - e^{-iat}}{2i}\right)$

$= \frac{1}{2i}\{\mathcal{L}(e^{iat}) - \mathcal{L}(e^{-iat})\}$

$= \frac{1}{2i}\left[\frac{1}{s-ia} - \frac{1}{s+ia}\right]$

$= \frac{1}{2i}\left[\frac{s+ia - s+ia}{s^2 - i^2a^2}\right]$

$$= \frac{1}{2i}\left[\frac{2ia}{s^2+a^2}\right] = \frac{a}{s^2+a^2}$$

Q. $\mathcal{L}[\cos at] = \frac{s}{s^2+a^2}$

A. $\cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2}$

$\therefore \mathcal{L}(\cos at) = \frac{1}{2}\{\mathcal{L}(e^{iat}) + \mathcal{L}(e^{-iat})\}$

$$= \frac{1}{2}\left[\frac{1}{s-ia} + \frac{1}{s+ia}\right]$$

$$= \frac{1}{2}\left[\frac{2s}{s^2+a^2}\right]$$

$$= \frac{s}{s^2+a^2}$$

Q. Find $\mathcal{L}(\sin^3 2t)$

A. $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$

$\therefore \sin^3\theta = \frac{3\sin\theta - \sin 3\theta}{4}$

$\sin^3 2t = \frac{3\sin 2t - \sin 6t}{4}$

$\therefore \mathcal{L}(\sin^3 2t) = \frac{3}{4}\mathcal{L}(\sin 2t) - \frac{1}{4}\mathcal{L}(\sin 6t)$

$= \frac{3}{4}\cdot\frac{2}{s^2+4} - \frac{1}{4}\frac{6}{s^2+36}$

$= \frac{48}{(s^2+4)(s^2+36)}$

Q. $\mathcal{L}(\sin 2t \sin 3t)$

A. $\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$

$\therefore \mathcal{L}(\sin 2t \sin 3t) = \mathcal{L}\left[\frac{1}{2}\{\cos t - \cos 5t\}\right]$

$= \frac{1}{2}\left\{\frac{s}{s^2+1} - \frac{s}{s^2+25}\right\}$

$= \frac{125}{(s^2+1)(s^2+25)}$

Q. Show that $\mathcal{L}\left(\frac{1}{\sqrt{\pi t}}\right) = \frac{1}{\sqrt{s}}$ if $\mathcal{L}\left(2\sqrt{\frac{t}{\pi}}\right) = \frac{1}{s^{3/2}}$

A. Let $F(t) = 2\sqrt{\frac{t}{\pi}}$

$F'(t) = \frac{1}{\sqrt{\pi t}}$

$\therefore \mathcal{L}\left(\frac{1}{\sqrt{\pi t}}\right) = \mathcal{L}[F'(t)] = s\mathcal{L}[F(t)] - F(0)$

$= s \cdot \frac{1}{s^{3/2}} - 0$

$= \frac{1}{\sqrt{s}}$

Q. $\mathcal{L}(t + t^2 + t^3)$

A $= \mathcal{L}(t) + \mathcal{L}(t^2) + \mathcal{L}(t^3)$

$= \frac{1!}{s^2} + \frac{2!}{s^3} + \frac{3!}{s^4}$

$\mathcal{L}[t^n] = \frac{n!}{s^{n+1}}$

Q. $\mathcal{L}(t\cosh at)$

A. $\mathcal{L}(\cosh at) = \frac{s}{s^2-a^2}$

RESULT:

$$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n}{ds}[F(s)] \; ; \; F(s) = \mathcal{L}[f(t)]$$

$\therefore \mathcal{L}(t\cosh at) = -1 \times \frac{d}{ds}\left(\frac{s}{s^2-a^2}\right)$

$= -1\left(\frac{-s^2-a^2}{(s^2-a^2)^2}\right)$

$= \frac{s^2+a^2}{(s^2-a^2)^2}$

Q. If $\mathcal{L}[f(t)] = F(s)$ then $\mathcal{L}\left\{\frac{1}{t}f(t)\right\} = \int_s^\infty F(s)\,ds$

A. $F(s) = \int_0^\infty e^{-st} f(t)\,dt$

$\int_s^\infty F(s)\,ds = \int_s^\infty \left(\int_0^\infty e^{-st} f(t)\,dt\right) ds$

$= \int_0^\infty f(t)\left[\int_s^\infty e^{-st}\,ds\right] dt$

$= \int_0^\infty f(t)\left[\frac{e^{-st}}{-t}\right]_s^\infty dt$

$= \int_0^\infty f(t)\left[0 - \frac{e^{-st}}{-t}\right] dt$

$= \int_0^\infty e^{-st}\left(\frac{1}{t}f(t)\right) dt = \mathcal{L}\left[\frac{1}{t}f(t)\right]$, hence proved.

Q. $\mathcal{L}\left(\frac{\sin 2t}{t}\right)$

A. $\mathcal{L}\left\{\frac{1}{t} f(t)\right\} = \int_s^\infty F(s)\,ds$

$\therefore F(s) = \mathcal{L}(\sin 2t)$

$= \frac{2}{s^2+4}$

$\therefore \mathcal{L}\left(\frac{1}{t} f(t)\right) = \int_s^\infty \frac{2}{s^2+4}\,ds$

$= 2\left[\tan^{-1}\left(\frac{s}{2}\right)\right]_s^\infty$

$= 2\left[\frac{\pi}{2} - \tan^{-1}\left(\frac{s}{2}\right)\right]$

$= \pi - 2\tan^{-1}\left(\frac{s}{2}\right)$

Q. $\mathcal{L}\left[t\, e^{-t} \sin 2t\right]$

A. $\mathcal{L}\left(t^n f(t)\right) = (-1)^n \frac{d^n}{ds}\left[F(s)\right]$

$\therefore \mathcal{L}\left[t\, e^{-t} \sin 2t\right) = -1 \frac{d}{ds} F(e^{-t} \sin 2t)$

Shifting Theorem:

$\mathcal{L}\left[e^{at} f(t)\right] = F(s-a)$; $F(s) = \mathcal{L}(f(t))$

$\therefore F(e^{-t} \sin 2t) = F(s+1) = \frac{2}{(s+1)^2+4}$

$$\therefore \mathcal{L}(te^{-t}\sin 2t) = -1\frac{d}{ds}\left(\frac{2}{(s+1)^2+4}\right)$$

$$= \frac{4(s+1)}{((s+1)^2+4)^2}$$

Table

$f(t)$	$\mathcal{L}(f(t))$
1	$\frac{1}{s}$
t	$\frac{1}{s^2}$
t^n $(n=0,1,2,..)$	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\cos at$	$\frac{s}{s^2+a^2}$
$\sin at$	$\frac{a}{s^2+a^2}$
$\cosh at$	$\frac{s}{s^2-a^2}$
$\sinh at$	$\frac{a}{s^2-a^2}$
$e^{at}\cos bt$	$\frac{s-a}{(s-a)^2+b^2}$
$e^{at}\sin bt$	$\frac{b}{(s-a)^2+b^2}$

shifting property

Inverse Laplace Transform

$$f(t) = \mathcal{L}^{-1}[F(s)]$$

Q. $\mathcal{L}^{-1}\left(\frac{1}{s^2-25}\right) = \mathcal{L}^{-1}\left(\frac{1}{s^2-5^2}\right) = \mathcal{L}^{-1}\left[\frac{1}{5}\ \frac{5}{s^2-5^2}\right)$

$$= \frac{1}{5}\mathcal{L}^{-1}\left[\frac{5}{s^2-5^2}\right] = \frac{1}{5}\sinh at$$

Q. $\mathcal{L}^{-1}\left[\frac{s^2+3s+8}{s^3}\right]$

A. $= \mathcal{L}^{-1}\left(\frac{1}{s} + \frac{3}{s^2} + \frac{8}{s^3}\right)$

$$= \mathcal{L}^{-1}\left(\frac{1}{s}\right) + \mathcal{L}^{-1}\left(\frac{3}{s^2}\right) + \mathcal{L}^{-1}\left(\frac{8}{s^3}\right)$$

$$= 1 + 3t + \frac{8}{2!}\mathcal{L}^{-1}\left(\frac{2!}{s^3}\right)$$

$$= 1 + 3t + 4t^2$$

Q. $\mathcal{L}^{-1}\left[\frac{1}{(s+1)(s+2)}\right]$ use partial fractions

A. $\frac{1}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$

$$1 = A(s+2) + B(s+1)$$

$$As + Bs = 0s$$

$$2A + B = 1$$

$$2A - A = 1$$

$$A = 1 \qquad B = -1$$

$$\therefore a = \frac{1}{s+1} - \frac{1}{s+2}$$

$$= \mathcal{L}^{-1}\left(\frac{1}{s+1}\right) - \mathcal{L}^{-1}\left(\frac{1}{s+2}\right)$$

$$= e^{-t} - e^{-2t}$$

Q. $\mathcal{L}^{-1}\left[\frac{3}{(s-2)^3}\right)$

A. $\mathcal{L}^{-1}\left[\frac{n!}{s^{n+1}}\right] = t^n$

$$\mathcal{L}^{-1}\left(\frac{1}{s^{n+1}}\right) = \frac{t^n}{n!}$$

$$\therefore Q = 3\mathcal{L}^{-1}\left(\frac{1}{(s-2)^3}\right)$$

$$= 3\, e^{2t}\mathcal{L}^{-1}\left(\frac{1}{s^3}\right)$$

$$= 3e^{2t}\frac{t^2}{2!}$$

$$= \frac{3}{2}e^{2t}t^2.$$

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Shifting Property:

$$\mathcal{L}(e^{at}f(t)) = F(s-a)$$

RESULTS:

1) $\mathcal{L}^{-1}\left[\frac{F(s)}{s}\right] = \int_0^t \mathcal{L}^{-1}[F(s)]\,dt = \int_0^t f(t)\,dt$

2) First Shifting Property.

If $\mathcal{L}^{-1}[F(s)] = f(t)$, then

$$\mathcal{L}^{-1}[F(s+a)] = e^{-at}\,\mathcal{L}^{-1}[F(s)]$$

3) Second shifting property.

$\mathcal{L}^{-1}[e^{-at}F(s)] = f(t-a)\cdot u(t-a)$; $u(t-a)$ is unit step functio

Q. $\mathcal{L}^{-1}\left[\frac{1}{s(s+a)}\right]$

$A. = \frac{1}{s}\cdot\frac{1}{s+a}$

$F(s) = \frac{1}{s+a}$

$\therefore \mathcal{L}^{-1}[F(s)] = e^{-at}$

$\therefore Q. = \int_0^t e^{-at}\,dt$

$= \left.\frac{e^{-at}}{-a}\right|_0^t$

$= -\frac{e^{-at}}{a} + 1 = \frac{1}{a}(1 - e^{-at})$

2. $\mathcal{L}^{-1}\ \frac{1}{s(s^2+1)}$

$$= F(s) = \frac{1}{s^2+1}$$

$\therefore f(t) = \sin t$

orpartial fractions

$$\frac{1}{s(s^2+1)} = \frac{A}{s} + \frac{Bs+C}{s^2+1}$$

$$\therefore a. = \int_0^t \sin t\, dt$$

$$= -\cos t \Big|_0^t$$

$$= -\cos t + 1$$

$$= 1 - \cos t$$

① Solve $y'' - y = t$; $y(0) = 1$, $y'(0) = 1$ using Laplace Transform

A. $y'' - y = t$

taking $\mathcal{L}$

$$\mathcal{L}(y'') - \mathcal{L}(y) = \mathcal{L}(t)$$

$$s^2\mathcal{L}(y) - sy(0) - y'(0) - \mathcal{L}(y) = \frac{1}{s^2}$$

$$\mathcal{L}(y)(s^2-1) - s - 1 = \frac{1}{s^2}$$

$$\mathcal{L}(y) = \mathcal{L}^{-1}\left(\frac{1}{s^2} + s + 1\right)\frac{1}{s^2-1}$$

$$\mathcal{L}(y) = \frac{s+1}{s^2-1} + \frac{1}{s^2(s^2-1)}$$

$$\mathcal{L}(y) = \frac{s+1}{(s+1)(s-1)} + \frac{1}{s^2(s^2-1)}$$

$$y = \mathcal{L}^{-1}\left[\frac{1}{s-1} + \frac{1}{s^2(s^2-1)}\right]$$

$$= \mathcal{L}^{-1}\left[\frac{1}{s-1} + \frac{1}{s^2-1} - \frac{1}{s^2}\right]$$

$$= e^t + \sinh t - t$$

---

2) $y'' + y' + 9y = 0$, $y(0) = 0.16$, $y'(0) = 0$

A. $s^2\mathcal{L}(y) - sy(0) - y'(0) + s\mathcal{L}(y) - y(0) + 9\mathcal{L}(y) = 0$

$$\mathcal{L}(y)[s^2+s+9) - 0.16s - 0.16 = 0$$

$$\mathcal{L}(y)(s^2+s+9) = 0.16(s+1)$$

$$y = \mathcal{L}^{-1}\left(\frac{0.16(s+1)}{s^2+s+9}\right) = \frac{0.16s}{s^2+s+9} + \frac{0.16}{s^2+s+9}$$

$$= \mathcal{L}^{-1}\left(\frac{0.16(s+\frac{1}{2}+\frac{1}{2})}{(s+\frac{1}{2})^2+\frac{35}{4}}\right)$$

$$= 0.16\left[\mathcal{L}^{-1}\left(\frac{s+\frac{1}{2}}{(s+\frac{1}{2})^2+\left(\sqrt{\frac{35}{4}}\right)^2}\right) + \mathcal{L}^{-1}\left(\frac{\frac{1}{2}}{(s+\frac{1}{2})^2+\left(\sqrt{\frac{35}{4}}\right)^2}\right)\right]$$

$\frac{s+a}{(s+a)^2+b^2}$ $\qquad$ $\frac{a}{s^2+a^2}$

$$= 0.16\left[\mathcal{L}^{-1}\left(\frac{s+\frac{1}{2}}{(s+\frac{1}{2})^2+\sqrt{\frac{35}{4}}^2}\right) + \frac{1\sqrt{4}}{2\sqrt{34}}\left(\frac{\frac{\sqrt{35}}{\sqrt{4}}}{(s+\frac{1}{2})^2+\sqrt{\frac{35}{4}}^2}\right)\right]$$

- $\mathcal{L}[e^{at}f(t)] = F(s-a)$

$$= 0.16\left[e^{-0.5t} \cos\sqrt{\frac{35}{4}}t + \frac{1}{\sqrt{35}} e^{-0.5t} \sin\frac{\sqrt{35}}{2}t \right)$$

$$= 0.16 e^{-0.5t}\left[\cos 2.96t + 0.027 \sin 2.96t \right]$$

3) $y'' - 4y' + 3y = 6t - 8 ; \quad y(0) = 0, y'(0) = 0$

$$= s^2 \mathcal{L}(y) - s y(0) - y'(0) - 4s\mathcal{L}(y) - 4y(0) + 3\mathcal{L}(y) = 6\mathcal{L}(t) - 8\mathcal{L}(1)$$

$$\mathcal{L}(y)[s^2 - 4s + 3] - 0 - 0 = \frac{6}{s^2} - \frac{8}{s}$$

$$\mathcal{L}(y) = \frac{6}{s^2(s^2 - 4s + 3)} - \frac{8}{s(s^2 - 4s + 3)}$$

$$= \frac{6}{s^2(s-1)(s-3)} - \frac{8}{s(s-1)(s-3)}$$

$$= \frac{6 - 8s}{s^2(s-1)(s-3)} -$$

$$\frac{6 - 8s}{s^2(s-1)(s-3)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-1} + \frac{D}{s-3}$$

$$6 - 8s = As(s-1)(s-3) + B(s-1)(s-3) + Cs^2(s-3) + Ds^2(s-1)$$

$s = 0$

$$6 = 3B$$

$$B = 2$$

$s = 1$

$$-2 = -2C \qquad C = 1$$

$s=3$

$$-18 = 18D$$

$$D = -1$$

$s=1$

$$14 = -8A + 8B - 4C - 2D$$

$$14 = -8A + 16 - 4 + 2$$

$$14 = -8A + 14$$

$$-8A = 0$$

$$A = 0$$

$$\therefore \mathcal{L}(y) = \frac{2}{s^2} + \frac{1}{s-1} + -\frac{1}{s-3}$$

$$\therefore y = 2\mathcal{L}^{-1}\left(\frac{1}{s^2}\right) + \mathcal{L}^{-1}\left(\frac{1}{s-1}\right) - \mathcal{L}^{-1}\left(\frac{1}{s-3}\right)$$

$$= 2t + e^t + e^{3t}$$

Q. System of ODE using Laplace Transform.

$$y_1' = a_{11} y_1 + a_{12} y_2 \quad \text{— (1)}$$

$$y_2' = a_{21} y_1 + a_{22} y_2 \quad \text{— (2)}$$

Taking Laplace transform on both sides,

$$\mathcal{L}(y_1') = a_{11}\mathcal{L}(y_1) + a_{12}\mathcal{L}(y_2)$$

$$\mathcal{L}(y_2') = a_{21}\mathcal{L}(y_1) + a_{22}\mathcal{L}(y_2)$$

i.e. $$s\mathcal{L}(y_1) - y_1(0) = a_{11}\mathcal{L}(y_1) + a_{12}\mathcal{L}(y_2)$$

$$s\mathcal{L}(y_2) - y_2(0) = a_{21}\mathcal{L}(y_1) + a_{22}\mathcal{L}(y_2)$$

2 eq. in 2 variables.

Solve normally.

Then take inverse Laplace Transform.

Q. $y_1' = -y_1 + 4y_2$

$y_2' = 3y_1 - 2y_2$; $y_1(0) = 3$; $y_2(0) = 4$

A. $s\mathcal{L}(y_1) - 3 = -\mathcal{L}(y_1) + 4\mathcal{L}(y_2)$ —①

$s\mathcal{L}(y_2) - 4 = 3\mathcal{L}(y_1) - 2\mathcal{L}(y_2)$ —②

①* $(s+1)\mathcal{L}(y_1) - 4\mathcal{L}(y_2) = 3$ —①

$(s+2)\mathcal{L}(y_2) - 3\mathcal{L}(y_1) = 4$ —②

① × 3

2 × (s+1)

$3(s+1)\mathcal{L}(y_1) - 12\mathcal{L}(y_2) = 9$

(+) $-3(s+1)(s+2) + (s+1)(s+2)\mathcal{L}(y_2) = 4(s+1)$

$(s+5)(s-2)\mathcal{L}(y_2) = 9 + 4(s+1)$

$\therefore \mathcal{L}(y_2) = \frac{9 + 4(s+1)}{(s+5)(s-2)} = \frac{4s+13}{(s+5)(s-2)}$

$$y_2 = \mathcal{L}^{-1}\left\{\frac{4s+13}{(s+5)(s-2}\right\}$$

by partial fractions,

$$\frac{4s+13}{(s+5)(s-2)} = \frac{A}{s+5} + \frac{B}{s-2}$$

$$4s+13 = A(s-2) + B(s+5)$$

$s=2$

$$8+13 = 21 = 7B$$

$$B = 3$$

$s=-5$

$$-7 = -7A \Rightarrow A = 1$$

$$\therefore y_2 = \mathcal{L}^{-1}\left(\frac{1}{s+5} + \frac{3}{s-2}\right)$$

$$y_2 = e^{-5t} + 3e^{2t}$$

From (1), $(s+1)\mathcal{L}(y_1) = 3 + 4\mathcal{L}(y_2)$

$$\mathcal{L}(y_1) = \frac{3}{s+1} + \frac{4(5+13)}{(s+1)(s+5)(s-2)}$$

OR

$$y_2' = -5e^{-5t} + 6e^{2t} = 3y_1 - 2e^{-5t} + 6e^{2t}$$